

Task 1. Look and Say Sequence

Available marks: 12

The *Look and Say Sequence* was proposed and studied by mathematician John H Conway in 1987 (it has some peculiar properties beyond the scope of this task). It starts with a single digit, say 1, and each element thereafter describes the structure of the previous element, thus:

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1
11      (the previous element comprises one 1)
21      (the previous element comprises two 1s)
1211    (the previous element comprises one 2 and one 1)
111221  (the previous element comprises one 1, one 2 and two 1s)
312211  (the previous element comprises three 1s, two 2s and one 1)
13112221 (the previous element comprises one 3, one 1, two 2s and two 1s)
...

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Your tasks

You have related two tasks:

- A. Write a program that can generate the look and say sequence starting at any given digit, and ending with the first element that contains more than 60 digits. Each element is to be shown on a separate line for clarity.
- B. You will find that apart from the starting digit, the elements are all permutations of the digits {1,2,3}. For the final element identified in Part A, find and display the longest non-overlapping repeated substring embedded within the element. For example, the longest repeated substring in "231131122211311323" is "11311" (underlined here, you just need to display the string).

Assessment

When you believe you have completed this task, show the judges the result of running the program with a starting digit that they specify.

7 marks are awarded for the correct sequence and **5 marks** for correctly identifying the longest repeated substring.

Source: Wolfram Mathworld

mathworld.wolfram.com/LookandSaySequence.html